Table S2 - Data extracted from the studies evaluating the performance of CT NAAT-based POCTs

Study	First Author	Publication Year	Location	Test evaluated	Specimen	Reference assay	confirm assay for discrepant results	Sample size	Population Prevalence (%)	Reported sensitivity (n/n)	Reported specificity (n/n)	PPV	NPV	DOR
1	Goldenberg SD	2012	UK	GeneXpert® CT/NG (Cepheid, Sunnyvale, CA)	Rectal swab	Hologic Aptima assay	NA	409 males	10.5	86.0% (37/43)	99.2% (363/366)	92.5%	98.4%	746
2	Gaydos CA	2013	USA	Cepheid GeneXpert CT/NG (Xpert) assay	Vaginal swab	AC2 (APTIMA Combo 2)/ BDPT (Becton,Dickinson ProbeTec)	NA	1713 females	4.6	98.7% (78/79)	99.4% (1624/1634)	88.6%	99.9%	12667
				Cepheid GeneXpert CT/NG (Xpert) assay	Cervical swabs	AC2 (APTIMA Combo 2)/ BDPT (Becton,Dickinson ProbeTec)	NA	1710 females	4.6	97.4% (76/78)	99.6% (1625/1632)	91.6%	99.9%	8821
				Cepheid GeneXpert CT/NG (Xpert) assay	Urine	AC2 (APTIMA Combo 2)/ BDPT (Becton,Dickinson ProbeTec)	NA	1718 females	4.8	97.6% (80/82)	99.8% (1633/1636)	96.4%	99.9%	21773

				Cepheid GeneXpert CT/NG (Xpert) assay	Urine	AC2 (APTIMA Combo 2)/ BDPT (Becton,Dickinson ProbeTec)	NA	1386 males	5.8	97.5% (79/81)	99.9% (1304/1305)	98.7%	99.8%	51508
3	Geiger R	2016	USA	The GeneXpert® CT/NG (Cepheid, Sunnyvale, CA)	Pharyngeal swab	APTIMA® system (Hologic, San Diego, CA)	NA	144 males	1.4	50.0% (1/2)	100.0% (142/142)	100.0%	99.3%	71
				The GeneXpert® CT/NG (Cepheid, Sunnyvale, CA)	Rectal swab	APTIMA® system (Hologic, San Diego, CA)	NA	141 males	12.1	88.2% (15/17)	100.0% (124/124)	100.0%	98.4%	1544
4	Bristow CC	2017	USA	XpertW CT/NG (Cepheid, Sunnyvale, CA)	Rectal swab	APTIMA® system (Hologic, San Diego, CA)	NA	393 males	12.5	85.7% (42/49)	99.4% (342/344)	95.5%	98.0%	1026
5	Cosentino LA	2017	USA	Cepheid Xpert CT/NG assay (Xpert)	Rectal swab	APTIMA® system (Hologic, San Diego, CA)	Aptima CT	399 (224male and 175 female)	13.5	94.4% (51/54)	98.3% (339/345)	89.5%	99.1%	961
				Cepheid Xpert CT/NG assay (Xpert)	Pharyngeal swab	APTIMA® system (Hologic, San Diego, CA)	NA	394 (224male and 175 female,5 invalid)	2.0	100.0% (8/8)	99.5% (384/386)	80%	100%	2615
				Cepheid Xpert CT/NG assay (Xpert)	Urine	APTIMA® system (Hologic, San Diego, CA)	NA	224 (male)	4.5	90.0% (9/10)	100% (214/214)	100%	99.5%	2717

				Cepheid Xpert CT/NG assay (Xpert)	Vaginal swabs	APTIMA® system (Hologic, San Diego, CA)	NA	170 (female)	10.0	58.8% (10/17)	99.3% (152/153)	90.9%	95.6%	217
6	Wilson SP	2017	USA	Xpert CT/NG Assay (Cepheid, Sunnyvale, CA)	Urine	Aptima Combo 2 Assay (Gen-Probe Incorporated, San Diego, CA)	NA	50 males	20.0	80% (8/10)	97.5% (39/40)	88.9%	95.1%	156
				Xpert CT/NG Assay (Cepheid, Sunnyvale, CA)	Urine	Aptima Combo 2 Assay (Gen-Probe Incorporated, San Diego, CA)	NA	1112 females	11.2	80.8% (101/125)	99.2% (979/987)	92.7%	97.6%	515
7	Dize L	2018	USA	Cepheid GeneXpert CT/NG assay	Rectal swab	Aptima Combo 2 Assay (Gen-Probe Incorporated, San Diego, CA)	NA	401 males and females	5.5	95.5% (21/22)	99.7% (378/379)	95.5%	99.7%	7938
8	Causer LM	2018	Australia	GeneXpert CT/NG assay	Self-collected urine or self-collected vaginal swab	NAAT	retest with Cobas 4800 or in-house *OmpA* CT assay	2486(male and female)	8.5	98.6% (209/212)	99.5% (2263/2274)	95.0%	99.9%	14332

9	Badman SG	2018	Papua New Guinea	Xpert CT/NG Test	Rectal swab	Cobas 4800 CT/NG test (Roche Molecular Diagnostics, Pleasanton, CA, USA)	NA	250 FSW	48.0	96.7% (116/120)	97.7% (127/130)	97.5%	96.9%	1228
				Xpert CT/NG Test	Rectal swab	Cobas 4800 CT/NG test (Roche Molecular Diagnostics, Pleasanton, CA, USA)	NA	76 MSM and transgender women	38.2	96.6% (28/29)	89.4% (42/47)	84.8%	97.7%	235
10	Harding-Esch EM	2018	UK	The Atlas Genetics io® platform	vaginal swab	CE-marked Becton Dickinson (BD) ProbeTec ™ Qx CT/GC assay (Oxford, UK)	Cepheid Xpert CT/NG	709 female	7.2	96.1% (49/51)	97.7% (643/658)	76.6%	99.7%	1050
11	Widdice LE	2018	USA	The Atlas Genetics io® platform	vaginal swabs	Aptima Combo 2TM	Cepheid Xpert CT/NG	284 female	10.9	83.9% (26/31)	98.8% (250/253)	89.7%	98.0%	433
12	Harding-Esch EM	2018	UK	The TwistDx RPA CT/ NG assay	urine	CT/NG nucleic acid amplification test (NAAT)	NA	392 female	8.9	94.3% (33/35)	99.7% (356/357)	97.1%	99.4%	5874
				The TwistDx RPA CT/ NG assay	vaginal swab	CT/NG nucleic acid amplification test (NAAT)	NA	395	7.1	96.4% (27/28)	100.0% (367/367)	100.0%	99.7%	13475

				The TwistDx RPA CT/ NG assay	urine	CT/NG nucleic acid amplification test (NAAT)	NA	395 male	7.3	100.0% (29/29)	100.0% (366/366)	100.0%	100.0%	43247
13	Sachdev D	2018	India	an in-house-developed beacon-based PCR diagnostic assay kit	cervical swabs	FTD Urethritis basic NAAT (qPCR) kit	NA	482 female	9.5	95.8% (46/48)	100% (434/434)	100.0%	99.5%	16163
14	Lehmusvuor i A	2010	Finland	assay based on homogenous PCR utilizing GenomEra concept(new assay)	urine(male or female not reported)	BD ProbeTec CT	BD ProbeTec CT	148 (male or female not reported)	50	98.7% (73/74)	97.3% (72/74)	97.3%	98.6%	2628
15	Pearce DM	2011	USA	Velox™ technology (new assay)	genital swab samples(male or female not reported)	BD ProbeTec CT	NA	102 (male or female not reported)	51	96.2% (50/52)	98.0% (49/50)	98.0%	96.1%	1225
16	Dean D	2012	USA	a microfluidic assay that simultaneousl y interrogates nine Ct loci(new assay)	cervical swabs	Roche-Amplicor NAAT	microfluidic sanger sequencing	263 female	54	91.5% (141/154)	100.0% (121/121)	100.0%	90.3%	2547
17	Melendez JH	2013	USA	the 16S rRNA assay(new assay)	vaginal swabs	ProbeTec or Gen-Probe	NA	257 female	17.5	75.5% (34/45)	92.9% (197/212)	69.4%	94.7%	41
				cryptic plasmid assays(new assay)	vaginal swabs	ProbeTec or Gen-Probe	NA	257 female	17.5	82.2% (37/45)	92.9% (197/212)	71.2%	96.1%	61

18	Van Der Pol B	2017	USA	The BD Max CT/GC/TV (MAX) assay	vaginal swabs	Becton Dickinson BD Qx assays or Hologic AC2	NA	1836 female	7.7	99.3% (140/141)	98.6% (1672/1695)	85.9%	99.9%	10177
				The BD Max CT/GC/TV (MAX) assay	Endocervic al swabs	Becton Dickinson BD Qx assays or Hologic AC2	NA	1831 female	7.5	95.7% (132/138)	99.2% (1680/1693)	91.0%	99.6%	2843
				The BD Max CT/GC/TV (MAX) assay	Female urine	Becton Dickinson BD Qx assays or Hologic AC2	NA	1849 female	7.7	91.5% (130/142)	99.5% (1699/1707)	94.2%	99.3%	2301
				The BD Max CT/GC/TV (MAX) assay	Male urine	Becton Dickinson BD Qx assays or Hologic AC2	NA	830 male	21.8	96.1% (174/181)	99.4% (645/649)	97.8%	98.9%	4008
19	Krõlov K	2014	Estonia	Recombinase polymerase amplification (RPA)-based POC	urine samples(51 females and 19 males)	Roche Cobas Amplicor C. trachomatis test (HoffmanLa Roche, Basel, Switzerland)	NA	70 patients (51 females and 19 males)	17.1	83.3% (10/12)	100.0% (58/58)	100.0%	96.7%	491
20	Miyazaki N	2016	Japan	GENECUBE	cervical swabs	APTIMA" Combo2 chlamydia/ gonorrhoeae, Hologic Japan, Inc.)	NA	95 female	22.1	90.5% (19/21)	98.6% (73/74)	95.0%	97.3%	694
21	Somboo nna N	2018	Thailand	LAMP-AuNP	cervical swabs	NAAT	NA	130 female	18.5	95.8% (23/24)	99.1% (105/106)	95.8%	99.1%	2415

22	Van Der Pol B	2020	USA	the binx health ioCT/NG assay (binx health)	vaginal swabs	NAAT	NA	1523 female	8.5	96.1% (124/129)	99.1% (1381/1394)	90.5%	99.6%	2635
				the binx health ioCT/NG assay (binx health)	urine	NAAT	NA	922 male	13	92.5% (111/120)	99.3% (796/802)	94.9%	98.9%	1636
23	Morris SR	2020	USA	the Visby Medical Sexual Health Test	vaginal swabs	the BD ProbeTec CT/GC Qx Amplifed DNA Assay	NA	1457 female	8.6	97.6% (122/125)	98.3% (1310/1332)	84.7%	99.8%	2422
24	Ye X	2021	China	the all-in-one microfluidic nucleic acid diagnosis system	swabs(urethra or vagina)	NAAT	NA	48(male or female not reported)	58.3	100.0%(28/28)	100.0%(20/20)	100.0%	100.0%	2337
25	Inoue T	2021	Japan	Rapid multiplex PCR assay	Cervical swabs	NAAT	NA	194 female	19.1	84.1% (37/44)	100.0% (150/150)	100.0%	95.5%	793
				Rapid multiplex PCR assay	first-void male urine	NAAT	NA	118 male	34.7	95.3% (41/43)	96.0% (72/75)	93.2%	97.3%	492